

Geometry of Reasoning Trajectories in Recurrent-Depth Transformers

Presenter

Marina Igitkhanian

M1 Applied Mathematics & Statistics, Institut Polytechnique de Paris

MAIN IDEA

Looped LMs reuse the same Transformer layers to think longer internally without adding more parameters or generating a long chain of thought.

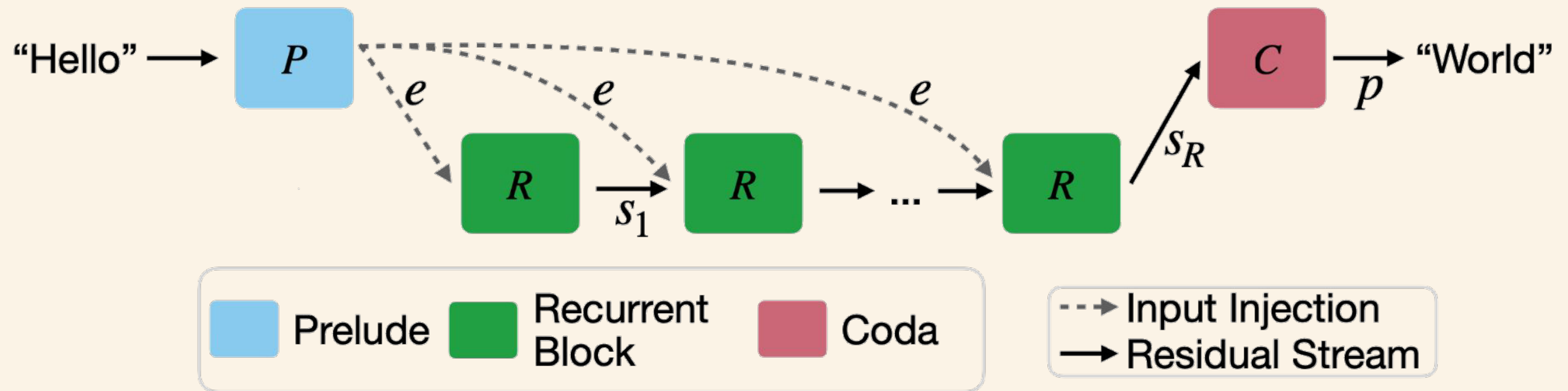


Figure 1. Recurrent-depth architecture adapted from Geiping et al.: a prelude encodes the input, a shared block iterates in latent space, and a coda produces the output.

THE GAP

Prior work often assumes that successful recurrent reasoning should converge and even proposes methods that explicitly enforce stable, fixed-point dynamics. This may overlook useful oscillatory or loop-like trajectories that still lead to correct predictions.

RESEARCH QUESTION

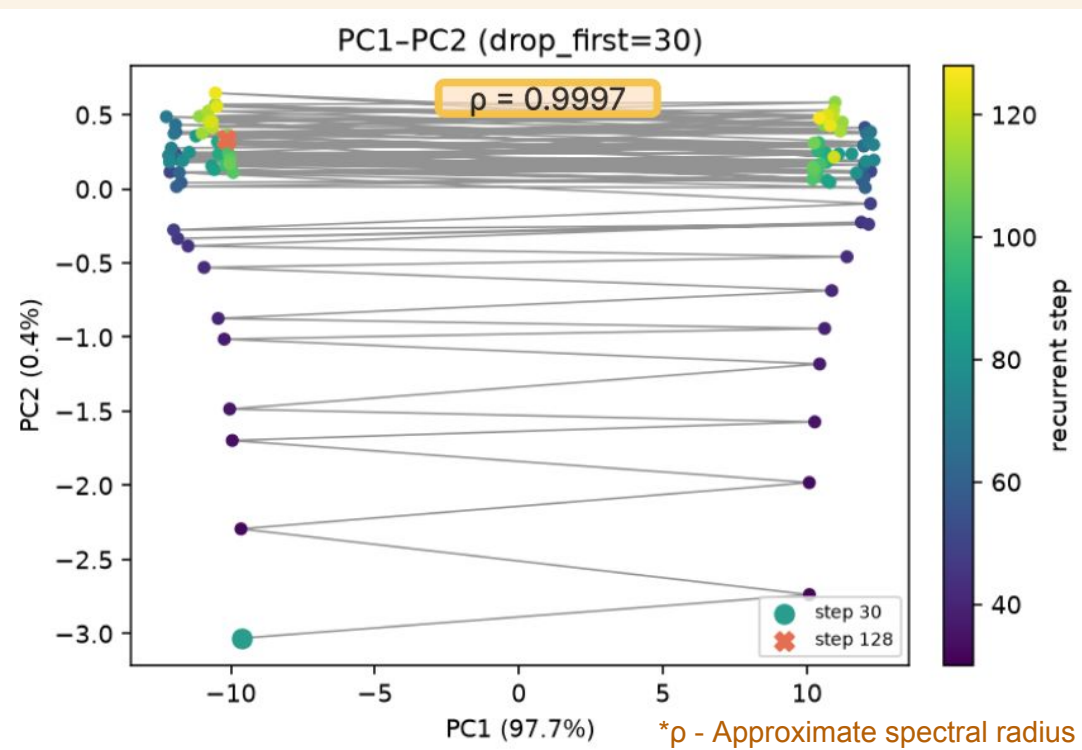
Do correct predictions require latent-state convergence, or can useful reasoning remain oscillatory or loop-like?

THE GOAL

Analyze Huginn's hidden-state trajectories and relate their geometry to intermediate predictions and final accuracy.

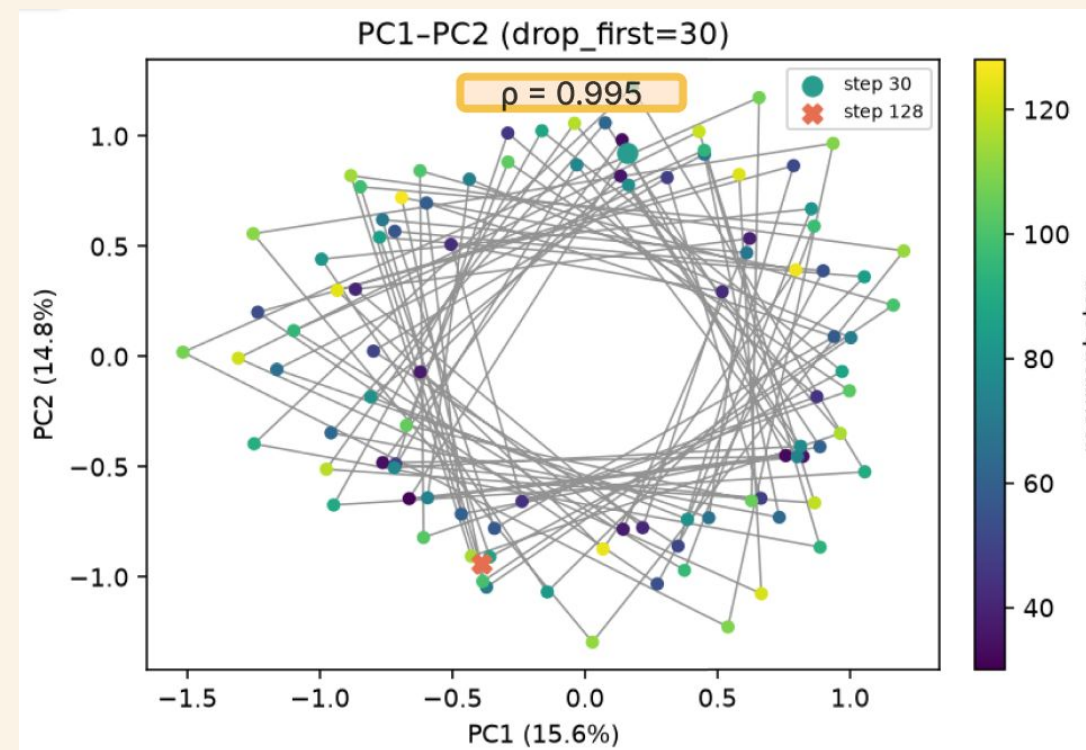
Results

Correct answers can follow very different latent trajectories



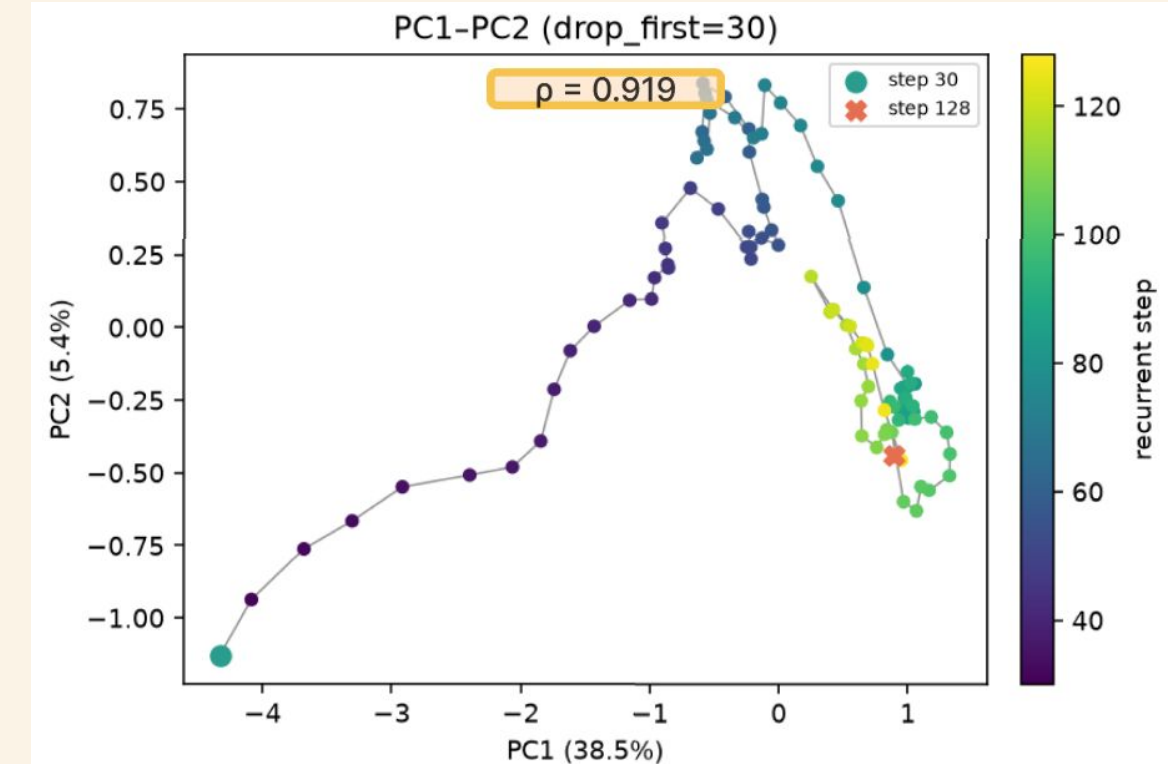
(a) Switching between two candidates

Trajectory oscillates sharply between correct and incorrect answer.



(b) Stable orbital motion

The model stays nearly 100% confident while the hidden state continues rotating.



(c) Damped oscillatory convergence

Correct and incorrect answers remain nearly equally probable.

Figure 2. PCA projections of answer-token trajectories over recurrent steps 30–128, showing distinct geometric shapes.

Results

The spectral-radius distributions are similar for trajectories leading to correct and incorrect answers.

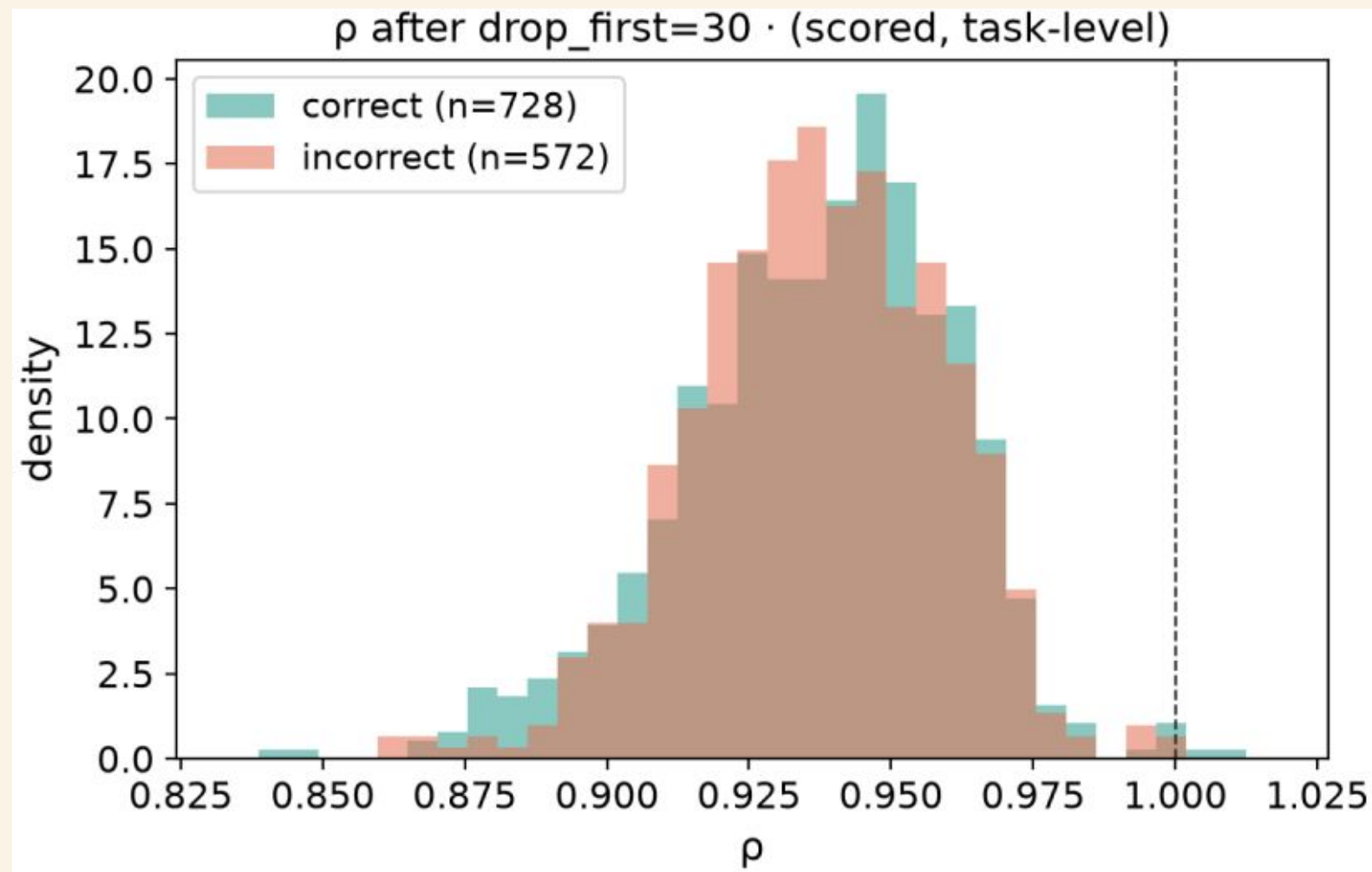


Figure 3. Spectral-radius distributions for correct and incorrect answers over recurrent steps 30–128 for tokens containing the final answer.

Methodology

We compare the spectral radius distributions for correct and incorrect answers.

Null hypothesis H0: the two distributions are the same.

Evaluation:

Histograms · Q–Q plots · Statistical tests

Conclusion

Do successful recurrent-depth language models need hidden-state convergence to reason correctly?

We analyzed **hidden-state trajectories in the Huginn** recurrent-depth Transformer, combining

- intermediate decoding
- PCA trajectory visualization
- spectral-radius analysis
- controlled reasoning tasks

Our experiments showed that:

- Correct answers can follow very different latent trajectories.
- Loop-like or oscillatory dynamics can still support confident predictions.
- Spectral radius alone does not clearly distinguish correct from incorrect answers.
- Latent geometry should be interpreted together with prediction behavior and task accuracy.

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